Thadoe Hein

AI Engineer

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PROFILE

AI Engineer with strong experience bridging research and production systems, specializing in MLOps, LLM systems, and scalable AI platforms. Proven ability to design end-to-end machine learning pipelines, production-grade RAG systems, and multi-agent workflows. Experienced in model lifecycle management, CI/CD pipelines, and deploying AI services with monitoring, logging, and performance optimization.

EXPERIENCE

SoeMindAI, Inc. 02/2026 – 05/2026

AI Developer (Freelance)

Improved **3 AI-powered products** across EdTech and recruitment, owning end-to-end development from architecture to deployment.

- Cut AI costs by ~30% via optimized prompt design and model selection. Automated the end-to-end presentation slide creation process, reducing a multi-hour workflow to just minutes.
- Improved a unified study platform with 10 learning tools including flashcards, summaries, mind maps, and practice generation. Improved answer quality on large documents through smarter retrieval, while reducing AI costs by cutting unnecessary token usage.
- Engineered a job prep platform focused on reducing user friction and resume tailoring time. The system integrates ATS compatibility, job matching, and interview prep into a single workflow, utilizing multi-source discovery.

Kiddee Lab 03/2026 – 05/2026

Curriculum Designer & AI Integrator Thailand

- **Designed and developed** comprehensive "Intro to AI" and "Machine Learning" curricula tailored for youth, and delivered hands-on training to upskill the professional teaching staff for scalable course delivery.
- **Spearheaded legacy system digitization** by consulting on and implementing an automated pipeline that leveraged Gemini AI vision models and advanced prompt engineering to extract and convert paper receipts into digital records.
- **Facilitated specialized AI workshops** across diverse demographics, teaching AI literacy and daily applications for students, and actionable AI-driven business strategies for adult professionals.

Language Understanding Laboratory (LU Lab) 06/2025 – 08/2025

Machine Learning Engineer Intern

- Fine-tuned transformer models (mT5) and conducted structured experimentation to improve EM/F1 scores.
- Built reproducible ML pipelines for training, evaluation, and data preprocessing using Hugging Face.
- Conducted quantitative research and experimentation for model optimization and validation.

Intelligent Systems Laboratory (ISL) - Assumption University of Thailand 01/2024 – 10/2025

AI Engineer Apprenticeship

- Designed and deployed production-grade RAG systems, reducing hallucinations and improving response reliability.
- Built end-to-end ML pipelines including data ingestion, embedding generation, retrieval, and model serving.
- Operationalized LLM applications via REST APIs, enabling integration into real-world systems and workflows.
- Implemented monitoring for model performance, latency, and token usage to ensure system reliability.
- Collaborated with cross-functional teams to integrate AI systems into existing applications.

EDUCATION

Assumption University of Thailand Bangkok, Thailand

Bachelor of Science in Computer Sciences

Cumulative GPA: 3.7/4.0

TECHNICAL SKILLS

AI Systems & Machine Learning

Large Language Models (LLMs), Retrieval Augmented Generation (RAG), AI Agents, NLP, Computer Vision, Embeddings, Predictive Modeling, OCR, Model Evaluation and Prompt Optimization

LLM Frameworks & AI Tooling

LangChain, LangGraph, OpenAI API, Hugging Face, Vector Databases (Qdrant), Multi-Agent Systems

Programming

Python, Node.js, Express, React, Next.js, typescript, SQL, REST APIs

Infrastructure & DevOps

Docker, CI/CD, AWS, GCP

LANGUAGES

English
Professional

Burmese
Native

Thai
Beginner

PROJECTS

Video Intelligence & Face Search Platform 10/2025

- Developed video person-search using face similarity and embedding-based retrieval
- Reduced video processing time by 90% through optimized detection, tracking, and embedding pipelines
- Achieved sub-second identity search across large datasets using vector search (Qdrant)
- Built scalable cloud architecture with AWS (Lambda, ECS, RDS) for distributed video processing
- Delivered a full-stack interface (Next.js) for querying and viewing timestamped results

RAG Platform for Document Intelligence 02/2025

- Built a document-based chatbot that answers questions using retrieval-augmented generation (RAG)
- Improved answer accuracy and grounding by integrating document retrieval into LLM responses
- Reduced response latency via automated document ingestion and indexing pipelines
- Enhanced answer quality with ranking strategies and prompt optimization

End-to-End OCR & Vision Pipeline for Receipt Data Localization 02/2025

- Built a computer vision pipeline to detect and localize text in receipt images by collecting and preparing a custom dataset of receipt data.
- Reduced manual annotation effort by 50+ hours by leveraging Azure AI Vision to automatically generate labeled data in JSON format, then converting it into YOLO-compatible annotations.Fine-tuned YOLOv11 on the generated dataset to optimize text detection performance.
- Achieved 0.93 text detection accuracy on Thai receipt image datasets through model training and evaluation.

LLM Fine Tuning for News Summarization 08/2025

- Improved summarization performance on a news dataset as measured by evaluation metrics during experimentation, by **fine tuning a FLAN T5 model using a reproducible training pipeline**.
- Enabled systematic model improvements as measured by incremental quality gains, by conducting **structured hyperparameter tuning and dataset preparation experiments**.